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DNA Sequencing and Technologies

Direct RNA Methylation Mapping in Cancer Transcriptomes

Team – 12

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INTRODUCTION

RNA carries a hidden layer of information

- RNA isn't just a passive copy of DNA chemical modifications (the epitranscriptome) control how mRNA is spliced, stabilized, and translated into protein.
- N6-methyladenosine (m6A), the most-studied modification, is increasingly linked to cancer growth, immune evasion, and drug resistance.
- Traditional detection methods (MeRIP-seq, antibody-based) are indirect, low-resolution, and destroy the RNA in the process.
- Oxford Nanopore Direct RNA Sequencing (DRS) reads native, intact RNA molecules through a nanopore, capturing modification signatures directly in the raw electrical current no chemical conversion, no amplification.
- This project is a pure dry-lab computational pipeline: applying validated ML tools to public cancer DRS data to map m6A patterns and identify differentially methylated transcripts.

A mature, benchmarked foundation

Author	Source	Contribution
Hendra et al.	*Nature Methods, 2022*	Introduced m6Anet, a multiple-instance-learning model for de novo m6A calling from DRS signal.
Pratanwanich et al.	*Nature Biotechnology, 2021*	Introduced xPore, a statistical model for detecting differential RNA modification between two conditions, validated on clinical cancer samples.
Maestri et al.	*Briefings in Bioinformatics, 2024*	Systematic benchmark of 14 m6A-detection tools on DRS data; tool choice significantly affects sensitivity/precision trade-offs.
Chen et al. (SG-NEx)	*Nature Methods, 2025*	Systematic benchmark of nanopore long-read RNA-seq across cancer cell lines — 113 samples, 13 cell lines/tissues (GIS).

Collectively, these establish DRS + ML modification calling as a mature, benchmarked approach — but most published applications stop at the 5–6 most common cancer cell lines.

RESEARCH GAP

Gaps this project addresses

01

Common-cancer bias

Public DRS m6A maps concentrate on lung (A549), liver (HepG2), breast (MCF7), colon (HCT116), leukemia (K562). Less-common cancers remain largely unanalyzed.

02

Stoichiometry gap

Traditional methods report only whether a site is methylated, not what fraction of molecules carry the mark. m6Anet enables ratio-level tumor vs. normal comparisons.

03

Tool complementarity gap

m6Anet (de novo, motif-restricted) and xPore (comparative, statistical) rely on different assumptions and are rarely used together on the same dataset.

04

Scope of this project

Apply the pipeline to an under-analyzed cell-line comparison, plus a first-pass, hypothesis-generating m6A profile of multiple myeloma (unreplicated single-patient samples).

PROBLEM STATEMENT

Why current m6A detection falls short

- Current m6A detection relies on antibody enrichment or bisulfite-like chemical conversion both indirect and low-resolution, requiring large RNA input.
- These methods give only a population-average, binary signal (“this gene can be methylated”) they cannot resolve modification at the level of individual RNA molecules.
- Cancer transcriptomes show abnormal methylation patterns, but most existing maps come from common cancers (lung, breast, colon, liver, leukemia); many cancer types remain uncharacterized at the epitranscriptomic level.
- **Problem: There is no reproducible, publicly-validated computational pipeline that leverages direct RNA sequencing to map differential m6A methylation in cancer at single-molecule resolution, cross-validated against independent clinical data.**

DATASET

SG-NEx — Singapore Nanopore Expression Project

Genome Institute of Singapore • Public on AWS Open Data (free, no-login): *registry.opendata.aws/sgnex*

Cell line	Type	Direct RNA replicates	m6A-validated
A549 (lung)	Common cancer	4	Yes
HepG2 (liver)	Common cancer	5	Yes
HCT116 (colon)	Common cancer	5	Yes
K562 (leukemia)	Common cancer	4	Yes
MCF7 (breast)	Common cancer	4	Yes
Myeloma (3 patients)	Rare, patient tissue	1 each	No
NCC24 (stomach)	Cancer, single sample	1	No

- Replicated cancer cell lines give statistically sound differential comparisons for xPore.
- Pre-processed, tool-ready files remove the biggest time cost (GPU basecalling) from the timeline.
- Myeloma patient samples offer a genuine first look at an under-studied cancer type — framed as exploratory.
- Public, secondary-use data requires no wet-lab work.

METHODOLOGY

Project architecture — four phases, eight stages

1 · SETUP

Feasibility & Data Audit

Confirm dataset, replicate counts, sign-off with guide.

Environment Setup

GPU compute, containerized tools, Nextflow/Snakemake skeleton.

2 · DATA PROCESSING

Data Acquisition & Basecalling

Download raw/pre-processed files; basecall with Dorado if needed.

Alignment & Signal Realignment

minimap2 alignment; f5c/nanopolish eventalign maps signal to reference.

3 · ANALYSIS

Modification Detection

m6Anet (de novo, stoichiometry) + xPore (differential modification).

Annotation & Differential Analysis

Transcript-region annotation, multiple-testing correction, pathway enrichment.

4 · VALIDATION & DELIVERY

Cross-Validation

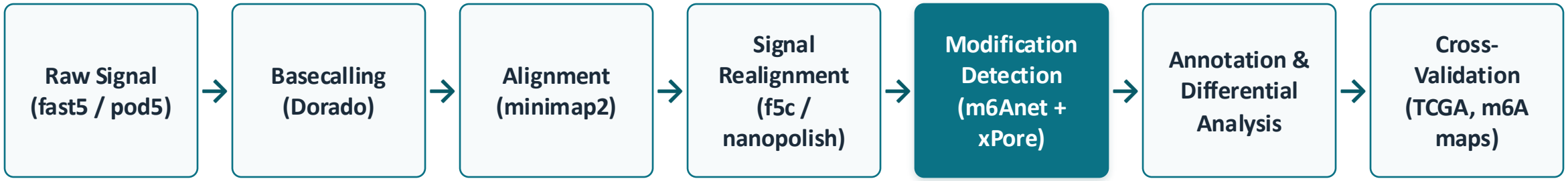
Check candidate genes against TCGA and independent MeRIP-seq/miCLIP maps.

Write-up & Reproducibility

Final report + public GitHub repo with workflow, containers, notebooks.

PROJECT PIPELINE

From raw signal to cross-validated results



TOOLS & ORCHESTRATION

Nextflow (nf-core/nanoseq) / Snakemake orchestration • Docker / Singularity containers • Git version control

Modification Detection is the analytical core — m6Anet and xPore run in parallel and are cross-checked against each other.

TECH STACK

Tools across the analysis stack

Layer	Tool
Orchestration	Nextflow (nf-core/nanoseq) or Snakemake
Basecalling	Dorado (GPU)
Alignment	minimap2
Signal realignment	f5c / nanopolish eventalign
Modification detection	m6Anet, xPore
Stats / annotation	R (GenomicRanges, DESeq2), Python (pandas, scikit-learn)
Validation	TCGA (GDC / UCSC Xena), independent MeRIP-seq / miCLIP
Compute	HPC / AWS / GCP GPU / Colab Pro

EXPECTED OUTCOMES

What this project delivers

Reusable pipeline

A documented, reusable Nextflow/Snakemake pipeline for epitranscriptomic DRS analysis.

Ranked transcript list

Ranked list of differentially methylated transcripts for the chosen cancer comparison.

Methods comparison

m6Anet vs. xPore behavior: precision/recall, motif bias, replicate sensitivity.

Myeloma m6A profile

Exploratory, hypothesis-generating first-pass m6A profile for multiple myeloma.

Sources

1 Hendra, C. et al. (2022). Detection of m6A from direct RNA sequencing using a multiple instance learning framework. Nature Methods.

2 Pratanwanich, N. et al. (2021). Identification of differential RNA modifications from nanopore direct RNA sequencing with xPore. Nature Biotechnology.

3 Maestri, S. et al. (2024). Benchmark of m6A detection tools on direct RNA sequencing data. Briefings in Bioinformatics.

4 Chen, Y., Davidson, N.M., Wan, Y.K. et al. (2025). A systematic benchmark of Nanopore long-read RNA sequencing for transcript-level analysis in human cell lines. Nature Methods, 22, 801–812.

5 GoekeLab/sg-nex-data and GoekeLab/m6anet GitHub repositories.

THANK YOU

Questions & Discussion

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